Week 4 Module

EASY LEVEL:

1. Problem Statement: Given a positive integer N, determine the total number of digits present in the number. Your task is to write a program that counts and prints the total number of digits.

```c
#include <stdio.h>

int main()
{
    long long n;
    int count = 0;
    scanf("%lld", &n);
    while (n > 0)
    {
        count++;
        n /= 10;
    }
    printf("%d\n", count);
     return 0;
}
```

Output

```
457305
6


=== Code Execution Successful ===
```

2. Problem Statement: Given a positive integer N, reverse its digits and print the reversed number. Your task is to write a program that reverses the given number.

```c
#include <stdio.h>

int main()
{
    long long n, reversed = 0;
    scanf("%lld", &n);
    while (n > 0)
    {
        int digit = n % 10;
        reversed = reversed * 10 + digit;
        n /= 10;
    }
    printf("%lld\n", reversed);
    return 0;
}
```

Output

```
1234
4321


=== Code Execution Successful ===
```

3. Problem Statement: Given a positive integer N, calculate the sum of all its digits. Your task is to write a program that computes and prints the sum of the digits.

```c
#include <stdio.h>

int main()
{
    long long n, sum = 0;
    scanf("%lld", &n);
    while (n > 0)
    {
        sum += n % 10;
        n /= 10;
    }
    printf("%lld\n", sum);
    return 0;
}
```

Output

```
5543
17


=== Code Execution Successful ===
```

4. Problem Statement: Given a positive integer N, count how many even digits and how many odd digits are present in the number. Your task is to write a program that prints the total count of even digits followed by the total count of odd digits.

```c
#include <stdio.h>

int main()
{
    long long n;
    int even_count = 0, odd_count = 0;
    scanf("%lld", &n);
    while (n > 0)
    {
        int digit = n % 10;
        if (digit % 2 == 0)
        {
            even_count++;
        } else
        {
            odd_count++;
        }
        n /= 10;
    }
    printf("%d %d\n", even_count, odd_count);
    return 0;
}
```

Output

```
223568
4 2


=== Code Execution Successful ===
```

5. Problem Statement: Given a positive integer N, find and print the largest digit present in the number. Your task is to write a program that determines the maximum digit.

```c
#include <stdio.h>

int main()
{
    long long n;
    int max_digit = 0;
    scanf("%lld", &n);
    while (n > 0)
    {
        int digit = n % 10;
        if (digit > max_digit)
        {
            max_digit = digit;
        }
        n /= 10;
    }
    printf("%d\n", max_digit);
    return 0;
}
```

Output

```
12658
8


=== Code Execution Successful ===
```

6. Problem Statement: Given two positive integers L and R, count how many prime numbers are present in the inclusive range [L, R]. A prime number is a natural number greater than 1 that has exactly two distinct positive divisors: 1 and itself. Your task is to write a program that counts and prints the total number of prime numbers in the given range.

```c
#include <stdio.h>
#include <stdbool.h>

bool is_prime(int num)
{
    if (num <= 1) return false;
    for (int i = 2; i * i <= num; i++)
    {
        if (num % i == 0) return false;
    }
    return true;
}

int main()
{
    int l, r, prime_count = 0;
    scanf("%d %d", &l, &r);

    for (int i = l; i <= r; i++)
    {
        if (is_prime(i))
        {
            prime_count++;
        }
    }
    printf("%d\n", prime_count);
    return 0;
}
```

Output

```
1 6
3


=== Code Execution Successful ===
```

7. Problem Statement: Given a positive integer N, find the sum of all its proper divisors. A proper divisor of a number is a positive divisor of the number excluding the number itself. Your task is to write a program that calculates and prints the sum of all proper divisors of N.

```
#include <stdio.h>

int main()
{
    int n;
    long long sum = 0;
    scanf("%d", &n);
    for (int i = 1; i <= n / 2; i++)
    {
        if (n % i == 0)
        {
            sum += i;
        }
    }
    printf("%lld\n", sum);
    return 0;
}
```

Output

```
8
7


=== Code Execution Successful ===
```

8. Problem Statement: Given a positive integer N, print all the factors of the number in ascending order. A factor of a number is an integer that divides the number completely without leaving any remainder. Your task is to write a program that prints all the factors of N.

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);
    for (int i = 1; i <= n; i++)
    {
        if (n % i == 0)
        {
            printf("%d ", i);
        }
    }
    printf("\n");
    return 0;
}
```

Output

```
10
1 2 5 10


=== Code Execution Successful ===
```

Handwritten notes

Week - 4 module

Handwritten Notes

Problem 1. Total number of digits

↳ Objective: Count the total digits in a positive integer (N)

↳ Logic: Use a (while (N > 0)) loop.

In each step, divide (N) by 10

(N = N / 10) to remove the last digit and increment a counter variable by 1.

Problem 2. Reverse the digits

↳ Objective: Reverse the digits of (N) and hande leading zeros.

↳ Logic: Extract the last digit using modulo ~~(N %~~

(N % 10).

Update the reversed variable using the formula:

reversed = (reversed × 10) + digit.

Finally, divide (N) by 10

Problem 3. Sum of all digits.

↳ Objective: Calculate the total sum of all digits in (N)

↳ Logic: Extract the last digit using (N % 10) and add it to a running 'sum' variable. then, divide (N) by 10 to move to the next digit.

Problem 4. Count Even and odd digits.

↳ Objective: Count how many digits in (N) are even and how many are odd.

↳ Logic: Extract a digit. If digit % 2 == 0, increment the even counter. otherwise, increment the odd counter. Divide (N) by 10 to continue.

Problem 5. Find the largest digit.

↳ Objective: Find and print the max^m digit present in (N)

↳ Logic: Initialize a 'max_digit' variable to 0. Extract digit ~~is~~ one by one. If the extracted digit is greater than 'max_digit', update max_digit. Divide (N) by 10.

Problem 6. Prime Numbers in a range [L, R]

↳ Objective: count all prime numbers b/w L to R inclusive

↳ Definition: A prime number is greater than 1 and has exactly two distinct divisor: 1 and itself.

↳ Logic: Use a loop from L to R. For each number, check if it is divisible by any integer up to its square root. If not, it is prime.

Problem 7. Sum of proper Divisors.

↳ Objective: Calculate the sum of all proper divisors of N

↳ Definition: A proper divisor is a positive divisor of a number, excluding the number itself.

↳ Logic: loop from '1 to N/2'. if 'N % i == 0' add 'i' to the sum variable.

Problem 8. Factors of a Number.

↳ Objective: Print all factors of 'N' in ascending order.

↳ Definition: A factor is an integer that divides 'N' completely, leaving no remainder.

↳ Logic: Loop from '1 to N'. Check if 'N % i == 0' If true, print 'i'.

--THE END--